

ROBERTO CORIA VARGAS

Computational Robotics Engineer | Full Stack Developer

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as Computational Robotics Engineer with 2+ years hands-on experience architecting end-to-end systems: from embedded firmware and industrial automation to full-stack web applications. Currently at Reyma del Sureste, designed and deployed solutions that cut operational processing time by 3+ hours. Expert across MERN stack, Computer Vision (OpenCV/YOLO), and automated data pipelines. Strategically integrate ML and autonomous agents into development workflows through disciplined prompt engineering and iterative refinement, moving beyond tool consumption toward deliberate system design.

PROFESSIONAL EXPERIENCE

Reyma del Sureste | Electromechanical Technician

March 2025 – Present

- Built pressurized-water leak detection system for molding line inspection. Eliminated manual disassembly, cutting inspection cycles from 3+ hours to checking internal passages without teardown. Extended design to handle acid wash integration for deep mold cleaning. System runs monthly saving ≈ 10 hours per cycle versus previous process.
- Perform preventive and corrective maintenance on high-speed thermoforming lines (5 ½ -16oz PP cups) and industrial mills. Diagnose mechanical failures and plan maintenance schedules to minimize production downtime.
- Technical documentation of internal processes through assembly and disassembly manuals for the mold area.

WINPRO | Systems Technician (Freelance)

2020 – 2024

- Diagnosed and fixed hardware/software issues on multiple client systems—RAM upgrades, SSD replacements, OS troubleshooting, malware removal. Maintained detailed documentation of problems and solutions for repeat client issues.
- Set up and configured Linux (Ubuntu) and Windows environments for clients. Handled OS-level updates, driver management, and system optimization.

PROJECTS

Personal | Zero Trail — Electric Vehicle Catalog

2026

- Live MERN app cataloging 3,000+ electric vehicles. Built full-stack: React frontend with MUI for responsive design, Node/Express API, MongoDB database. Users can filter by battery type, range, motor specs and return results in milliseconds. Deployed on Vercel, actively used by visitors. Code on GitHub with full documentation.

Personal | Mentec Labs — Desktop optimizer

2026

- Windows desktop optimizer built with Electron + React + TypeScript. Clears junk files, disables heavy background processes, manages startup apps. Real-world testing shows up to 15% performance boost on most systems. Published on GitHub with open source code and documentation.

UPY | Multi-task robot

2024

- Built robot using ROS with Publisher/Subscriber architecture (Python + C++). Raspberry Pi 4 ran the main controller coordinating sensors and motors via custom drivers.
- **Trained custom YOLO classifier on 1500+ images captured locally (streets, signs, pedestrians). Processed with Roboflow, achieved 85% accuracy. Integrated into robot's vision pipeline using OpenCV for real-time detection.**
- SolidWorks CAD design, 3D-printed chassis, remote access via VNC/SSH, dual manual/autonomous control via Tkinter interface.

UPY | Classification System with Machine Vision

2023

- **Wired 3 Siemens S7-1200 PLCs connected via PROFINET (TIA Portal), programmed part sorting logic by material (metal/plastic) and quality condition (good/defective).**
- COGNEX In-Sight vision system for quality validation; **Mitsubishi RV-4FR robotic arm (6 DOF)** for final part placement.
- Conveyor belt with 5 industrial sensors (fiber optic, inductive, capacitive, reflective, diffuse) and Siemens MicroMaster 420 inverter.

EDUCATION

Universidad Politécnica de Yucatán | *Eng. Computational Robotics Engineering*

Aug 2021 – Apr 2025

- GPA: **9.5 / 10** | State Robotics Tournament I6: Maze Solver, Task Robot, and Battle Robot categories.

English certification: ITEP Plus B2 — December 2024

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML5, CSS3, Java, TypeScript, Google App Script.

Robotics & embedded: ROS, ESP32, Raspberry Pi, Arduino, LoRa, UART, I2C, OpenCV, YOLO/Roboflow

Industrial automation: Siemens S7-1200 PLCs, TIA Portal, PROFINET, Mitsubishi RV-4FR, COGNEX In-Sight

Web development: React, Node.js, Express.js, MongoDB, REST API, MUI, Docker, Git

Tools: SolidWorks, LabVIEW, Linux (Ubuntu), Docker, VNC, SSH